



NORTH SYDNEY BOYS HIGH SCHOOL

2015 HSC ASSESSMENT TASK 1

Mathematics

General Instructions

- Working time – 55 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Ms Lee
- ☐ Mr Berry
- ☐ Ms Ziazaris
- ☐ Mr Zuber
- ☐ Mr Ireland
- ☐ Mr Lam
- ☐ Mr Lin

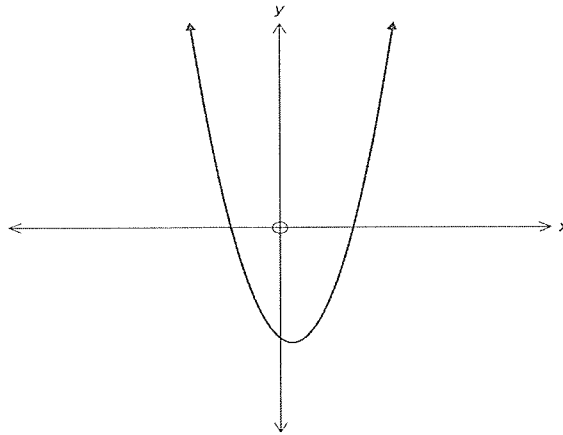
Student Number: _____

(To be used by the exam markers only.)

Question No	1-5	6	7	8	Total	Total %
Mark	$\frac{\quad}{5}$	$\frac{\quad}{26}$	$\frac{\quad}{9}$	$\frac{\quad}{11}$	$\frac{\quad}{51}$	$\frac{\quad}{100}$

QUESTION 1-5 MULTIPLE CHOICE (5 Marks)

1.



The graph above shows $y = ax^2 + bx + c$. Which of the following is true?

- (a) $a > 0$ & $\Delta > 0$ (b) $a > 0$ & $\Delta < 0$
 (c) $a < 0$ & $\Delta > 0$ (d) $a < 0$ & $\Delta < 0$

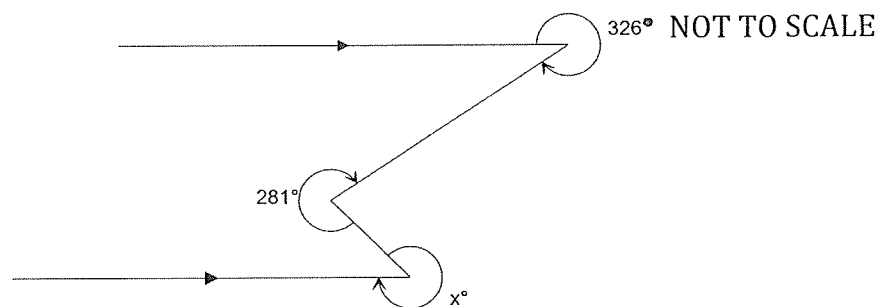
2. $P(x, y)$ moves such that P is equidistant from the line $y = -1$ and the point $(0, 1)$. What is the equation of the locus of P?

- (a) $y = \frac{x^2}{4}$ (b) $y = \frac{x^2}{2}$
 (c) $y = x^2$ (d) $y = 2x^2$

3. Which of the following is the length of the latus rectum of $y = x^2$?

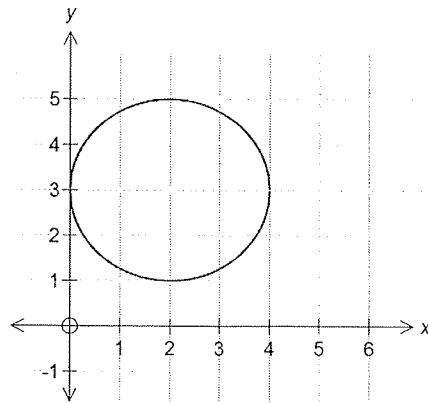
- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
 (c) 1 (d) 2

4. Calculate the value of the pronumeral x



- (a) 310° (b) 315°
 (c) 320° (d) 325°

5.



NOT TO SCALE

What is the equation of the figure shown above?

(a) $(x - 2)^2 + (y - 3)^2 = 2$ (b) $(x - 2)^2 + (y - 3)^2 = 4$

(c) $(x + 2)^2 + (y + 3)^2 = 2$ (d) $(x + 2)^2 + (y + 3)^2 = 4$

QUESTION 6 - QUADRATICS (26 Marks)

(a) (i) Sketch the graph of:

$$y = 2x^2 + 7x - 4$$

3

clearly showing the vertex and intercepts with the axes.

(ii) Hence or otherwise, solve:

$$2x^2 + 7x \leq 4$$

1

(b) Solve for x :

$$2x^2 - 8x + 1 = 0$$

2

(c) Solve for x :

$$9^x - 10 \times 3^x + 9 = 0$$

3

(d) For what values of k will the equation

$$x^2 - (k + 3)x + 4k = 0$$

have:

(i) Equal roots?

2

(ii) One root equal to 5?

1

(iii) Roots which are reciprocals of each other?

1

(iv) Roots of equal magnitude but opposite sign?

1

(v) Calculate the value of the roots in (iv).

1

- (e) If α and β are the roots of:

$$2x^2 + 5x - 1 = 0$$

Without solving the equation, find the values of:

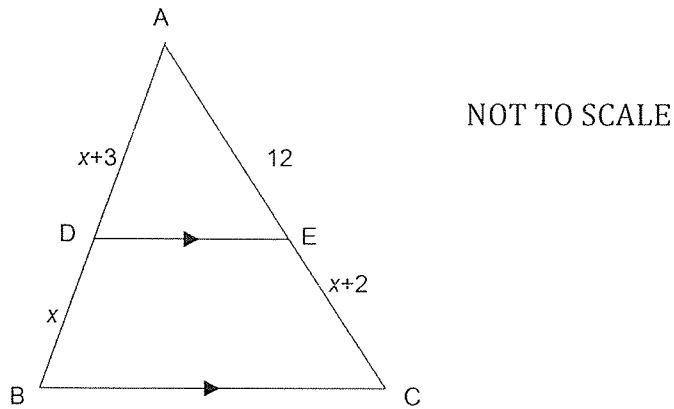
- | | |
|--|---|
| (i) $\alpha + \beta$ | 1 |
| (ii) $\alpha\beta$ | 1 |
| (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ | 2 |
| (iv) $\alpha^2 + \beta^2$ | 2 |
| (v) $ \alpha - \beta $ | 2 |
- (f) Express $2x^2 - 7x - 4$ in the form $a(x + 2)^2 + b(x + 2) + c$ 3

QUESTION 7 – LOCUS AND THE PARABOLA (9 Marks)

- (a) A point $P(x, y)$ moves so that it is always equidistant from the points $A(-3, 5)$ and $B(2, -1)$.
- | | |
|---|---|
| (i) Write the equation for the locus of P. | 2 |
| (ii) Sketch the equation of the locus of P on the number plane | 2 |
| (iii) Hence or otherwise, shade the region on the number plane where $PA \leq PB$ | 1 |
- (b) Sketch on the number plane:
- $$y^2 + 4y - 24x + 4 = 0$$
- clearly showing the focus, vertex and directrix 4

QUESTION 8 – PLANE GEOMETRY (11 Marks)

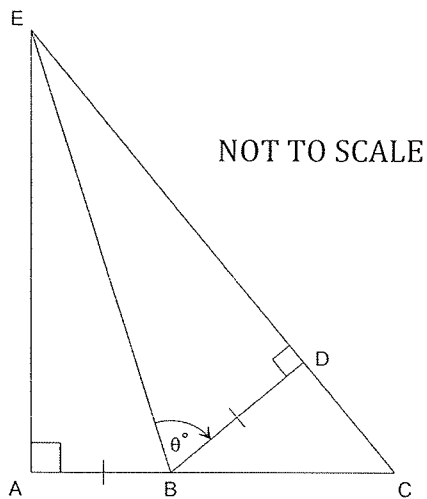
(a)



In the diagram above, find the value(s) of x .

3

- (b) In the diagram, $\triangle ABE$ is a right-angled triangle. The point B lies on AC and the point D lies on CE . Also $\angle BDE = 90^\circ$, $AB = BD$ and $\angle DBE = \theta$.



- (i) Show that $\triangle ABE \equiv \triangle DBE$

3

- (ii) Show that $\angle ACE = (2\theta - 90)^\circ$

2

- (iii) Show that $\triangle ACE \parallel \triangle DCB$

2

- (iv) Hence show that $\frac{EA}{AB} = \frac{CE}{CB}$

1

END OF EXAMINATION

Q1./ A

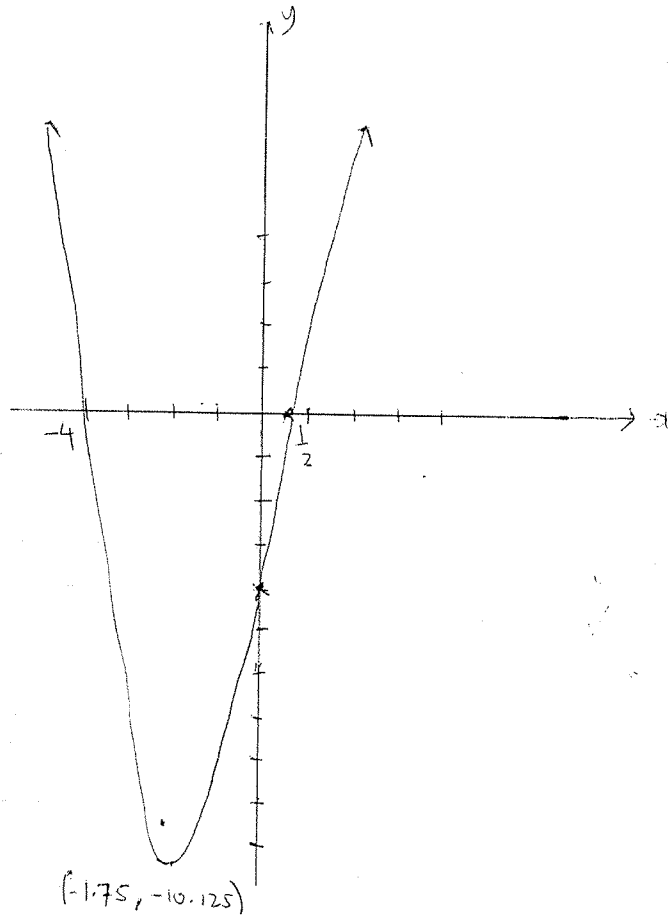
Q2./ A

Q3./ C

Q4./ B

Q5./ B

Q6 a1/



Q6 a2/ $-4 \leq x \leq \frac{1}{2}$

$$\begin{aligned} \text{Q6 b./} \quad x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{8 \pm \sqrt{64 - 4 \times 2 \times 1}}{2 \times 2} \\ &= \frac{8 \pm \sqrt{56}}{4} \\ &= \frac{8 \pm 2\sqrt{14}}{4} \\ &= \frac{4 \pm \sqrt{14}}{2} \end{aligned}$$

6 c/ let $u = 3^x$

$$u^2 - 10u + 9 = 0$$

$$(u - 9)(u - 1) = 0$$

$$\therefore u = 9 \quad \text{or} \quad u = 1$$

$$3^x = 9 \quad \text{or} \quad 3^x = 1$$

$$\therefore x = 2 \quad \text{or} \quad x = 0$$

$$(d)_i / \Delta = 0$$

$$b^2 - 4ac = 0$$

$$(-(k+3))^2 - 4 \times 1 \times 4k = 0$$

$$k^2 + 6k + 9 - 16k = 0$$

$$k^2 - 10k + 9 = 0$$

$$(k-9)(k-1) = 0$$

$$\therefore k = 1 \text{ or } k = 9$$

$$\text{dii} / 5^2 - (k+3) \times 5 + 4k = 0$$

$$25 - 5k - 15 + 4k = 0$$

$$\therefore -k + 10 = 0$$

$$\therefore k = 10$$

$$\text{diii} / \alpha \times \frac{1}{\alpha} = 1$$

$$\therefore \frac{4k}{1} = 1$$

$$\therefore k = \frac{1}{4}$$

$$(d) \text{iv} / \alpha + (-\alpha) = 0$$

$$\therefore \frac{k+3}{1} = 0$$

$$\therefore k = -3$$

$$\text{dv} / \alpha \times (-\alpha) = \frac{4k}{1}$$

$$-\alpha^2 = 4 \times (-3)$$

$$\alpha^2 = 12$$

$$\alpha = \pm \sqrt{12}$$

$$\therefore \text{roots are } \pm 2\sqrt{3}$$

$$\text{ei} / \alpha + \beta = -\frac{5}{2}$$

$$\text{eii} / \alpha \beta = -\frac{1}{2}$$

$$\text{eiii} / \frac{1}{\alpha} + \frac{1}{\beta}$$

$$= \frac{\alpha + \beta}{\alpha \beta}$$

$$= \frac{-\frac{5}{2}}{-\frac{1}{2}}$$

$$= 5$$

$$\text{ev/ } (\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$$

$$\therefore \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \left(-\frac{5}{2}\right)^2 - 2\left(-\frac{1}{2}\right)$$

$$= \frac{25}{4} + 1$$

$$= \frac{29}{4}$$

$$\text{ev/ } |\alpha - \beta| = \sqrt{(\alpha - \beta)^2}$$

$$= \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$$

$$= \sqrt{\left(-\frac{5}{2}\right)^2 - 4\left(-\frac{1}{2}\right)}$$

$$= \sqrt{\frac{25}{4} + 2}$$

$$= \sqrt{\frac{33}{4}}$$

$$= \frac{\sqrt{33}}{2}$$

$$\text{f/ } a(x+2)^2 + b(x+2) + c \equiv 2x^2 - 7x - 4$$

$$a(x^2 + 4x + 4) + bx + 2b + c \equiv 2x^2 - 7x - 4$$

$$ax^2 + 4ax + 4a + bx + 2b + c \equiv 2x^2 - 7x - 4$$

$$\therefore ax^2 + (4a + b)x + 4a + 2b + c \equiv 2x^2 - 7x - 4$$

$$\therefore \boxed{a = 2}$$

$$\therefore 4 \times 2 + b = -7$$

$$\boxed{b = -15}$$

$$4 \times 2 + 2 \times (-15) + c = -4$$

$$\therefore c = -4 - 8 + 30$$

$$\therefore \boxed{c = 18}$$

equating
co-efficients

$$7ai/ \sqrt{(x+3)^2 + (y-5)^2} = \sqrt{(x-2)^2 + (y+1)^2}$$

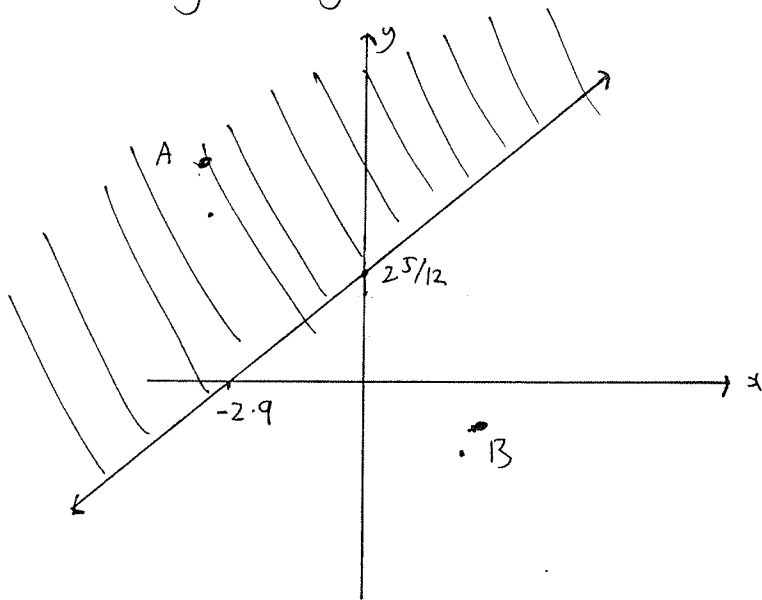
$$\therefore x^2 + 6x + 9 + y^2 - 10y + 25 = x^2 - 4x + 4 + y^2 + 2y + 1$$

$$\therefore 10x - 12y + 29 = 0$$

✓ some progress with
a correct method
✓ correct answer.

$$aii/ 12y = 10x + 29$$

$$\therefore y = \frac{5}{6}x + 2\frac{5}{12}$$

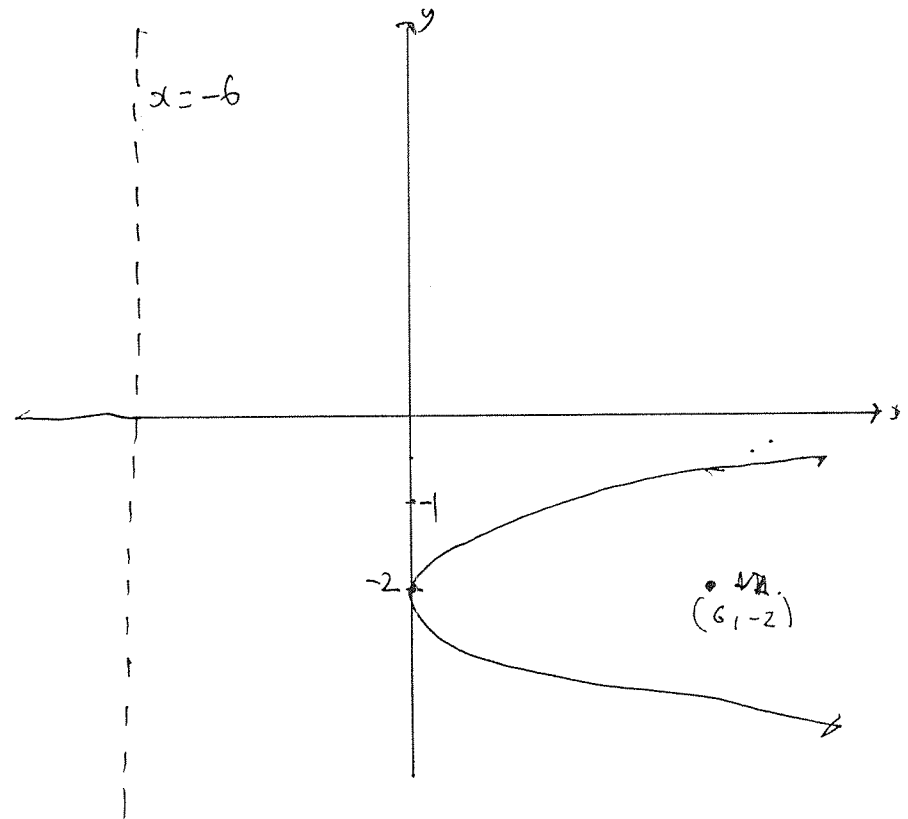


aiii/ region shaded on graph

$$b/ y^2 + 4y = 24x - 4$$

$$y^2 + 4y + 4 = 24x$$

$$(y+2)^2 = 4 \times 6(x-0)$$



8a/

$$\frac{x+3}{x} = \frac{12}{x+2} \quad (\text{ratio of intercepts})$$

$$(x+3)(x+2) = 12x$$

$$x^2 + 2x + 3x + 6 = 12x$$

$$x^2 + 5x + 6 = 12x$$

$$x^2 - 7x + 6 = 0$$

$$(x-6)(x-1) = 0$$

$$\therefore x = 6 \quad \text{or} \quad x = 1$$

8bi/ In ΔABE & ΔDBE

EB is common

$$\angle EAB = \angle EDB \quad (\text{given } - 90^\circ)$$

$$AB = DB \quad (\text{given})$$

$$\therefore \Delta ABE \cong \Delta DBE \quad (\text{RHS})$$

$$\text{bii/ } \angle ABE = \angle DBE \quad \left(\begin{array}{l} \text{corresponding angles in congruent} \\ \text{triangles } \Delta ABE \text{ \& } \Delta DBE \end{array} \right)$$

$$= \theta$$

$$\therefore \angle BDC + \angle BCD = \angle ABD \quad (\text{exterior angle of } \Delta BCD)$$

$$\therefore 90 + \angle BCD = 2\theta$$

$$\therefore \angle BCD = (2\theta - 90)^\circ$$

$$\therefore \angle ACE = (2\theta - 90^\circ)$$

(as $\angle BCD = \angle ACE$)

biii/ In ΔACE & ΔDCB

$$\angle ACE = \angle DCB \quad (\text{common to both})$$

$$\angle EAC = \angle BDC \quad (90^\circ - \text{given})$$

$$\therefore \Delta ACE \parallel \Delta DCB \quad (\text{equiangular})$$

$$\text{biv/ } \frac{EA}{DB} = \frac{CE}{CB} \quad \left(\begin{array}{l} \text{corresponding sides in similar} \\ \text{triangles } \Delta ACE \text{ \& } \Delta DCB \end{array} \right)$$

$$\therefore \frac{EA}{AB} = \frac{CE}{CB} \quad (\text{since } DB = AB - \text{given})$$

$$\text{show that } EA \times CB = AB \times CE$$